

Title : Time Domain Down Sampling

Category : Preprocessing

1- Objective :

The purpose of Time Domain Down Sampling is to reduce the temporal sampling rate of EEG signals by keeping fewer time points from the original signal. This helps decrease data size, reduce computational cost, and speed up later processing and analysis steps while preserving the important characteristics of the EEG signal.

2- how it works :

The method processes EEG data trial by trial. First, the user provides a downsampling factor, which determines how much the sampling rate will be reduced. For example, a factor of 2 reduces the sampling rate by half.

The algorithm groups the EEG data according to subject, session, and trial identifiers. For each trial, the EEG channels are converted into an MNE Raw object. The MNE resampling function is then applied to reduce the sampling frequency. During this process, an anti-aliasing filter is automatically used to minimize signal distortion and prevent frequency artifacts.

3- Example :

Suppose an EEG signal is recorded at a sampling rate of 512 Hz and the downsampling factor is set to 2.

Original sampling rate: 512 Hz

Downsampling factor: 2

New sampling rate: 256 Hz

If the original signal contains 10,240 time samples, the downsampled signal will contain approximately 5,120 samples.